Corn Simulation Data Analysis

Exploring yield and water use efficiency across genotypes, planting dates, and water management scenarios

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1) Data import and setup

Load necessary libraries and import the corn simulation data from the Excel file.

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.1
v lubridate 1.9.4     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(knitr)
library(stringr)
library(readxl)
# Load the corn simulation data
corn_sim <- read_excel(
  '/Users/lucasromano/Desktop/Crop Modeling/Corn SSM/ssm-model/SSM_iCrop_Maize_realbps.xlsm'
   sheet = "corn_sim")
```

2) Data cleaning and feature engineering

Cleaning and preparing the data.

```r
corn_sim_clean <- corn_sim %>%
  rename(YEAR = Pyear) %>%
  dplyr::select(-c(R2.BOT, R3.EAR, R4.ANT, R5.BSG, R6.TSG))  %>%
  dplyr::select(-NLF:-CNDNIT) %>%
  mutate(
    # Categorize water management from Management column
    water_mgmt = case_when(
      str_detect(Manag, "RFD") ~ "Rainfed",
      str_detect(Manag, "IRRI") ~ "Irrigated",
      TRUE ~ "Unknown"
    ),
    # Extract VPD response: Check vs LT (Limited Transpiration) genotypes
    vpd_response = case_when(
      str_detect(Crop, regex("check", ignore_case = TRUE)) ~ "Check",
      str_detect(Crop, regex("LT", ignore_case = TRUE)) ~ "LT"
    ),
    # Categorize planting date from scenario name
    Planting_date = case_when(
      str_detect(sName, "ELY") ~ "Early",
      str_detect(sName, "MID") ~ "Medium",
      str_detect(sName, "LTE") ~ "Late",
      TRUE ~ "Unknown"
    ),
    # Calculate water loss (runoff + evaporation + transpiration + drainage)
    WLOSS = CRUNOF + CE + CTR + CDRAIN,
    # Convert grain yield to kg/ha (WGRN is in smaller units)
    Yield_kgha = WGRN * 10,
    # Calculate Water Use Efficiency (yield per unit transpiration)
    WUE = Yield_kgha / ET,
    # Assign state based on location
    State = case_when(
      Location %in% c("Marianna", "Jonesboro", "Stuttgart") ~ "Arkansas",
      Location %in% c("Norfolk", "Lincoln", "North Platte") ~ "Nebraska",
      TRUE ~ "Other"
    )
  )
```

3) Data subsets

Data subsets for rainfed and irrigated scenarios.

```
rainfed_data <- corn_sim_clean %>%
  filter(water_mgmt == "Rainfed")

irrigated_data <- corn_sim_clean %>%
  filter(water_mgmt == "Irrigated")
```

Data subsets for early, medium, and late planting dates.

```
early_planting <- corn_sim_clean %>%
  filter(Planting_date == "Early")
medium_planting <- corn_sim_clean %>%
  filter(Planting_date == "Medium")
late_planting <- corn_sim_clean %>%
  filter(Planting_date == "Late")
```

Data subsets for Check and LT genotypes.

```
check_genotypes <- corn_sim_clean %>%
  filter(vpd_response == "Check")
lt_genotypes <- corn_sim_clean %>%
  filter(vpd_response == "LT")
```

Data subsets for locations in Arkansas and Nebraska.

```
marianna_data <- corn_sim_clean %>%
  filter(Location == "Marianna")

jonesboro_data <- corn_sim_clean %>%
  filter(Location == "Jonesboro")

stuttgart_data <- corn_sim_clean %>%
  filter(Location == "Stuttgart")

norfolk_data <- corn_sim_clean %>%
  filter(Location == "Norfolk")

lincoln_data <- corn_sim_clean %>%
```

```r
  filter(Location == "Lincoln")

north_platte_data <- corn_sim_clean %>%
  filter(Location == "North Platte")
```

4) Exploratory summaries and baseline comparisons

Rainfed scenarios.

```r
# Summary statistics for yield and water use efficiency by water
# management and planting date
summary_stats <- corn_sim_clean %>%
  group_by(water_mgmt, Planting_date) %>%
  summarise(
    mean_yield = mean(Yield_kgha, na.rm = TRUE),
    sd_yield = sd(Yield_kgha, na.rm = TRUE),
    mean_wue = mean(WUE, na.rm = TRUE),
    sd_wue = sd(WUE, na.rm = TRUE)
  )
```

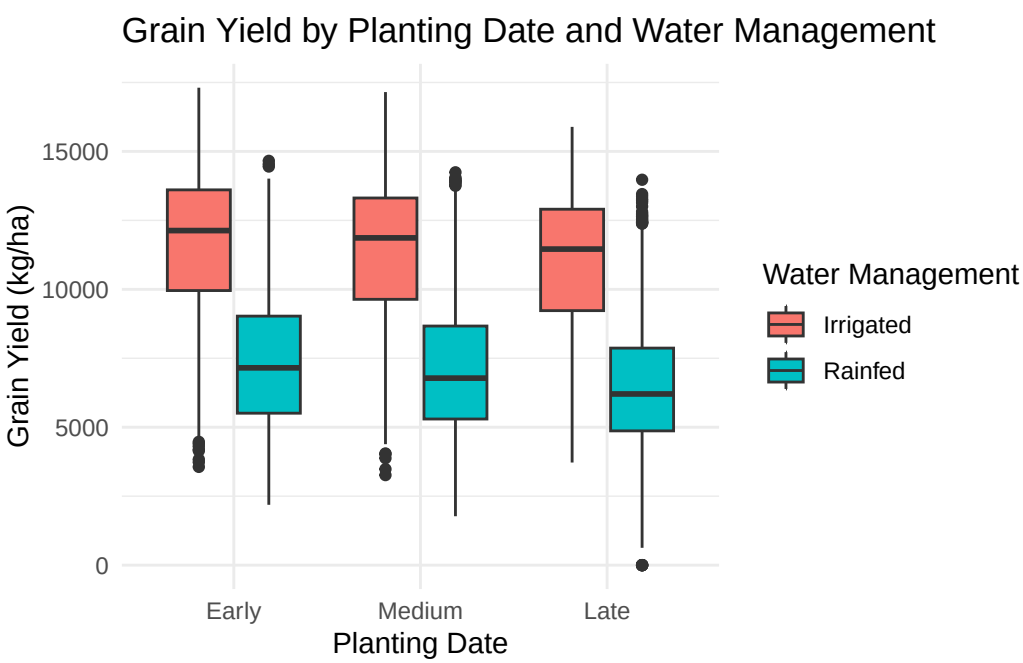
```
`summarise()` has grouped output by 'water_mgmt'. You can override using the
`.groups` argument.
```

```r
kable(
  summary_stats,
  caption = "Summary Statistics for Yield and Water Use Efficiency
  by Water Management and Planting Date")
```

Table 1: Summary Statistics for Yield and Water Use Efficiency by Water Management and Planting Date

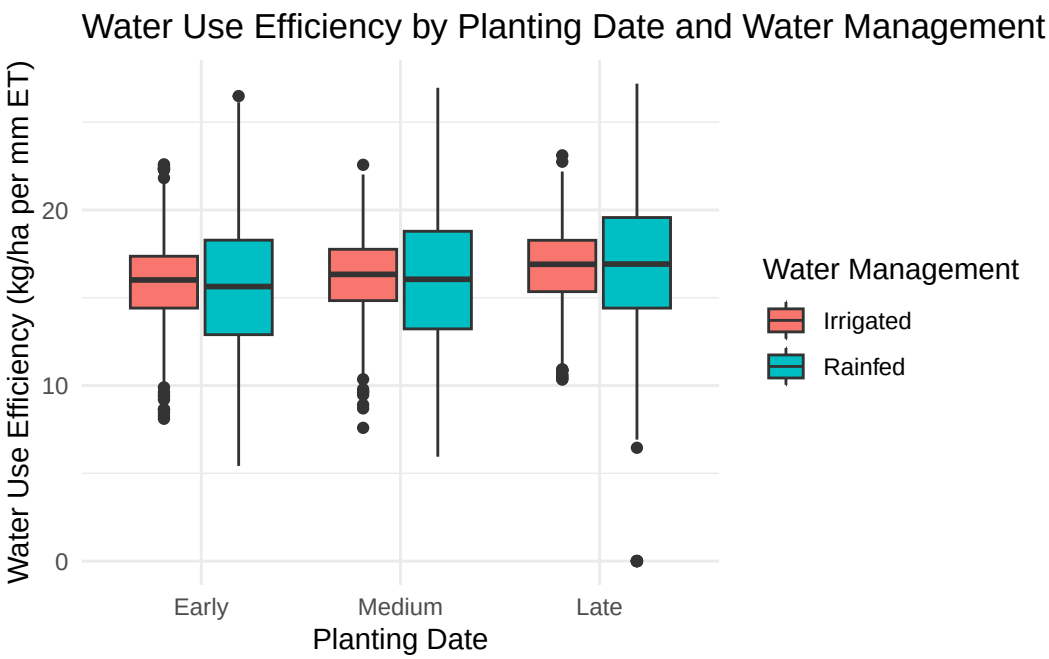
water_mgmt	Planting_date	mean_yield	sd_yield	mean_wue	sd_wue
Irrigated	Early	11504.392	2754.859	15.83278	2.357365
Irrigated	Late	10837.867	2690.758	16.74065	2.193579
Irrigated	Medium	11243.096	2728.346	16.23381	2.230904
Rainfed	Early	7374.020	2630.176	15.65764	3.808700
Rainfed	Late	6476.838	2443.027	16.84709	3.920416
Rainfed	Medium	7047.234	2569.860	16.02162	3.788677

```r
# Boxplot of yield by water management and planting date
ggplot(corn_sim_clean,
 aes(
  x = factor(
    Planting_date, levels = c("Early", "Medium", "Late")),
     y = Yield_kgha, fill = water_mgmt)) +
  geom_boxplot() +
  labs(title = "Grain Yield by Planting Date and Water Management",
       x = "Planting Date",
       y = "Grain Yield (kg/ha)",
       fill = "Water Management") +
  theme_minimal()
```

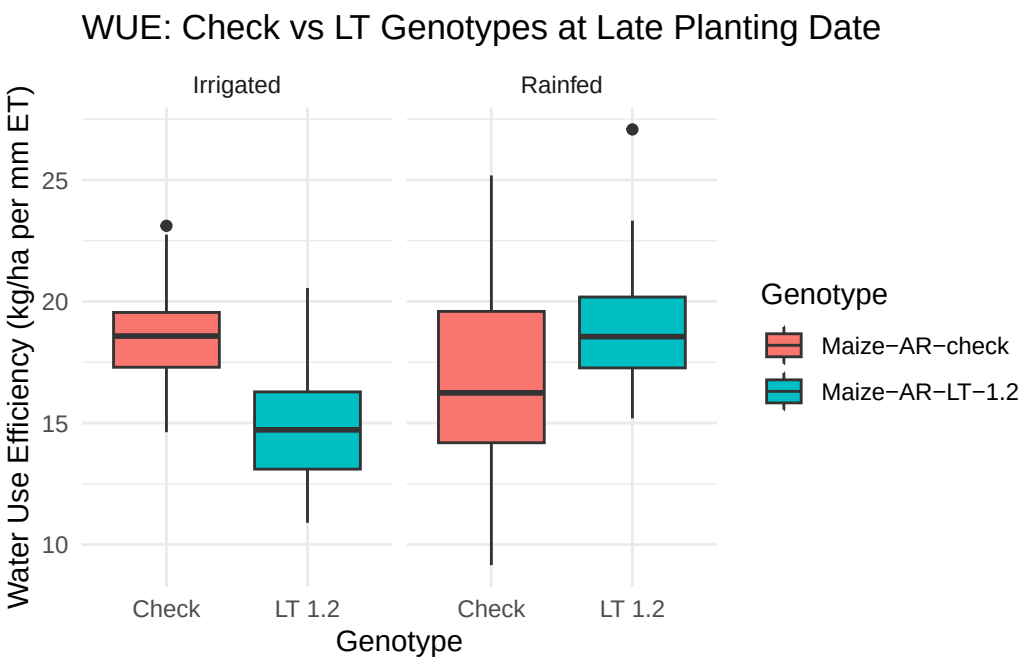



```r
# Boxplot of water use efficiency by water management and planting date
ggplot(
  corn_sim_clean, aes(
    x = factor(Planting_date, levels = c("Early", "Medium", "Late")),
    y = WUE, fill = water_mgmt)) +
  geom_boxplot() +
  labs(title = "Water Use Efficiency by Planting Date and Water Management",
       x = "Planting Date",
       y = "Water Use Efficiency (kg/ha per mm ET)",
```

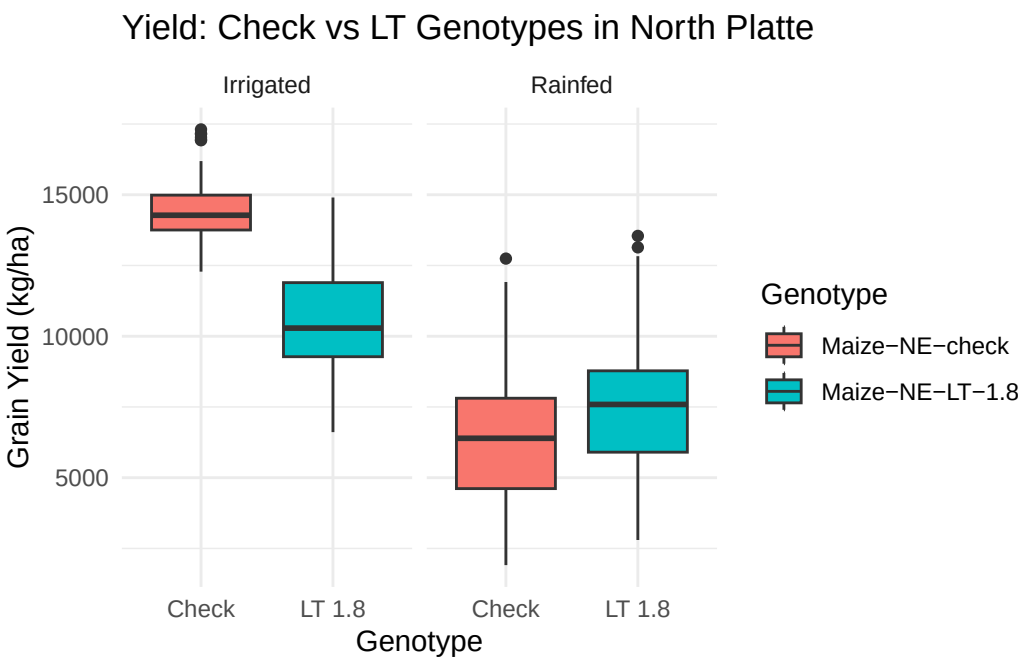
```
       fill = "Water Management") +
  theme_minimal()
```




```
# Compare WUE between Check and LT genotypes at Late planting date
corn_sim_clean %>%
  filter(Planting_date == "Late" &
  Crop %in% c("Maize-AR-check","Maize-AR-LT-1.2")) %>%
  ggplot(aes(x = Crop, y = WUE, fill = Crop)) +
  geom_boxplot() +
  facet_wrap(~ water_mgmt) +
  scale_x_discrete(
    labels = c("Maize-AR-check" = "Check", "Maize-AR-LT-1.2" = "LT 1.2")) +
  labs(title = "WUE: Check vs LT Genotypes at Late Planting Date",
       x = "Genotype",
       y = "Water Use Efficiency (kg/ha per mm ET)",
       fill = "Genotype") +
  theme_minimal()
```




```
# Analyze yield differences between Check and LT genotypes in North Platte
corn_sim_clean %>%
  filter(
    Crop %in% c("Maize-NE-check","Maize-NE-LT-1.8") &
    Location %in% c("North Platte")) %>%
  ggplot(aes(x = Crop, y = Yield_kgha, fill = Crop)) +
  geom_boxplot() +
  facet_wrap(~ water_mgmt) +
  scale_x_discrete(
    labels = c("Maize-NE-check" = "Check", "Maize-NE-LT-1.8" = "LT 1.8")) +
  labs(title = "Yield: Check vs LT Genotypes in North Platte",
       x = "Genotype",
       y = "Grain Yield (kg/ha)",
       fill = "Genotype") +
  theme_minimal()
```



4.1) Rainfed scenarios

Average yields by location.

```
avg_yields_location <- rainfed_data %>%
  group_by(Location) %>%
  summarise(
    mean_yield = mean(Yield_kgha, na.rm = TRUE),
    sd_yield = sd(Yield_kgha, na.rm = TRUE),
    n = dplyr::n(),
    .groups = "drop"
  )
knitr::kable(avg_yields_location, caption = "Average Yields by Location (kg/ha)")
```

Table 2: Average Yields by Location (kg/ha)

Location	mean_yield	sd_yield	n
Jonesboro	7317.767	2583.466	540
Lincoln	6020.854	2376.941	540
Marianna	7926.127	2643.484	540
Norfolk	7174.826	2562.368	540

Location	mean_yield	sd_yield	n
North Platte	6655.226	2364.819	540
Stuttgart	6701.383	2493.962	540

Average yields by planting date.

```
avg_yields_planting <- rainfed_data %>%
mutate(Planting_date = factor(
  Planting_date, levels = c("Early", "Medium", "Late"))) %>%
  group_by(Planting_date) %>%
  summarise(
    mean_yield = mean(Yield_kgha, na.rm = TRUE),
    sd_yield = sd(Yield_kgha, na.rm = TRUE),
    n = dplyr::n(),
    .groups = "drop"
  )
knitr::kable(avg_yields_planting, caption = "Average Yields by Planting Date (kg/ha)")
```

Table 3: Average Yields by Planting Date (kg/ha)

Planting_date	mean_yield	sd_yield	n
Early	7374.020	2630.176	1080
Medium	7047.234	2569.860	1080
Late	6476.838	2443.027	1080

Average yields by genotypes.

```
avg_yields_genotypes_AR <- rainfed_data %>%
  filter(Location %in% c("Marianna", "Jonesboro", "Stuttgart")) %>%
  group_by(Crop) %>%
  summarise(
    mean_yield = mean(Yield_kgha, na.rm = TRUE),
    sd_yield = sd(Yield_kgha, na.rm = TRUE),
    n = dplyr::n(),
    .groups = "drop"
  )
knitr::kable(avg_yields_genotypes_AR, caption = "Average Yields by Genotypes
in Arkansas (Check vs LT) (kg/ha)")
```

Table 4: Average Yields by Genotypes in Arkansas (Check vs LT) (kg/ha)

Crop	mean_yield	sd_yield	n
Maize-AR-LT-1.2	6575.547	1195.471	270
Maize-AR-LT-1.8	7836.036	2576.699	270
Maize-AR-LT-2.1	7512.340	2763.528	270
Maize-AR-LT-2.3	7329.207	2795.502	270
Maize-AR-LT-3.7	7055.122	2824.873	270
Maize-AR-check	7582.304	2973.777	270

```
avg_yields_genotypes_NE <- rainfed_data %>%
  filter(Location %in% c("North Platte","Lincoln","Norfolk")) %>%
  group_by(Crop) %>%
  summarise(
    mean_yield = mean(Yield_kgha, na.rm = TRUE),
    sd_yield = sd(Yield_kgha, na.rm = TRUE),
    n = dplyr::n(),
    .groups = "drop"
  )
knitr::kable(avg_yields_genotypes_NE, caption = "Average Yields by Genotypes
in Nebraska (Check vs LT) (kg/ha)")
```

Table 5: Average Yields by Genotypes in Nebraska (Check vs LT) (kg/ha)

Crop	mean_yield	sd_yield	n
Maize-NE-LT-1.2	6241.548	1728.095	270
Maize-NE-LT-1.8	7187.963	2551.008	270
Maize-NE-LT-2.1	6829.933	2574.517	270
Maize-NE-LT-2.3	6633.588	2555.840	270
Maize-NE-LT-3.7	6224.312	2542.704	270
Maize-NE-check	6584.468	2689.429	270

5) Weather data processing

Process historical weather data (1990-2019) for growing season (DOY 115-250). Classify weather years as: Cool/Wet, Warm/Wet, Normal, Cool/Dry, Warm/Dry.

```
  # Function to process weather data for a location
  process_weather_data <- function(file_path, location_name) {
```

```r
  read_excel(file_path, sheet = "Sheet2") %>%
    filter(between(DOY, 115, 250), between(YEAR, 1990, 2019)) %>%
    mutate(MeanT = (TMAX + TMIN) / 2) %>%
    group_by(YEAR) %>%
    summarise(Rain = sum(RAIN, na.rm = TRUE),
              Meat = mean(MeanT, na.rm = TRUE),
              .groups = "drop") %>%
    mutate(
      Rain.avg = mean(Rain),
      Meat.avg = mean(Meat),
      Rain.sd = sd(Rain),
      Meat.sd = sd(Meat),
      Normal = ifelse(Rain < Rain.avg + Rain.sd * 0.5 & Rain > Rain.avg - Rain.sd * 0.5 &
                      Meat < Meat.avg + Meat.sd * 0.5 & Meat > Meat.avg - Meat.sd * 0.5,
                      "Yes", "No"),
      Rain.class = ifelse(Rain > Rain.avg, "Wet", "Dry"),
      Meat_class = ifelse(Meat > Meat.avg, "Warm", "Cool"),
      weather.class = factor(
        ifelse(Normal == "Yes", "Normal", paste0(Meat_class, "/", Rain.class)),
        levels = c("Cool/Wet", "Warm/Wet", "Normal", "Cool/Dry", "Warm/Dry")
      ),
      Location = location_name
    )
}

# Define file paths and locations
weather_files <- list(
  Jonesboro = '/Users/lucasromano/Desktop/Crop Modeling/Weather/AR-ARTJBR_Jonesboro.xlsx',
  Stuttgart = '/Users/lucasromano/Desktop/Crop Modeling/Weather/AR-AR6920_Stuttgart.xlsx',
  Marianna = '/Users/lucasromano/Desktop/Crop Modeling/Weather/AR-AR4638_Marianna.xlsx',
  Lincoln = '/Users/lucasromano/Desktop/Crop Modeling/Weather/NE_Lincoln.xlsx',
  `North Platte` = '/Users/lucasromano/Desktop/Crop Modeling/Weather/NE-NE6075_NorthPlate_
  Norfolk = '/Users/lucasromano/Desktop/Crop Modeling/Weather/NE_Norfolk.xlsx'
)

# Process all locations and bind together
RAIN_ALL <- map_df(names(weather_files), ~process_weather_data(weather_files[[.x]], .x))
```

6) Merge simulation and weather classes

Join weather classification (Cool/Wet, Warm/Dry, etc.) to each year's yield data.

```r
corn_sim_classes <- left_join(x = corn_sim_clean,y = RAIN_ALL, by = c("YEAR","Location"))
```

6.1) Figure helper

```r
emit_figure <- function(plot_obj,
                        caption,
                        save_path = NULL,
                        width = 9,
                        height = 4,
                        dpi = 300,
                        device = "tiff") {
  old_cap <- knitr::opts_current$get("fig.cap")
  on.exit(knitr::opts_current$set(fig.cap = old_cap), add = TRUE)

  knitr::opts_current$set(fig.cap = caption)
  print(plot_obj)

  if (!is.null(save_path)) {
    ggsave(
      filename = save_path,
      plot = plot_obj,
      width = width,
      height = height,
      dpi = dpi,
      device = device
    )
  }
}
```

7) Year-by-year yield advantage (LT-1.8 vs Check)

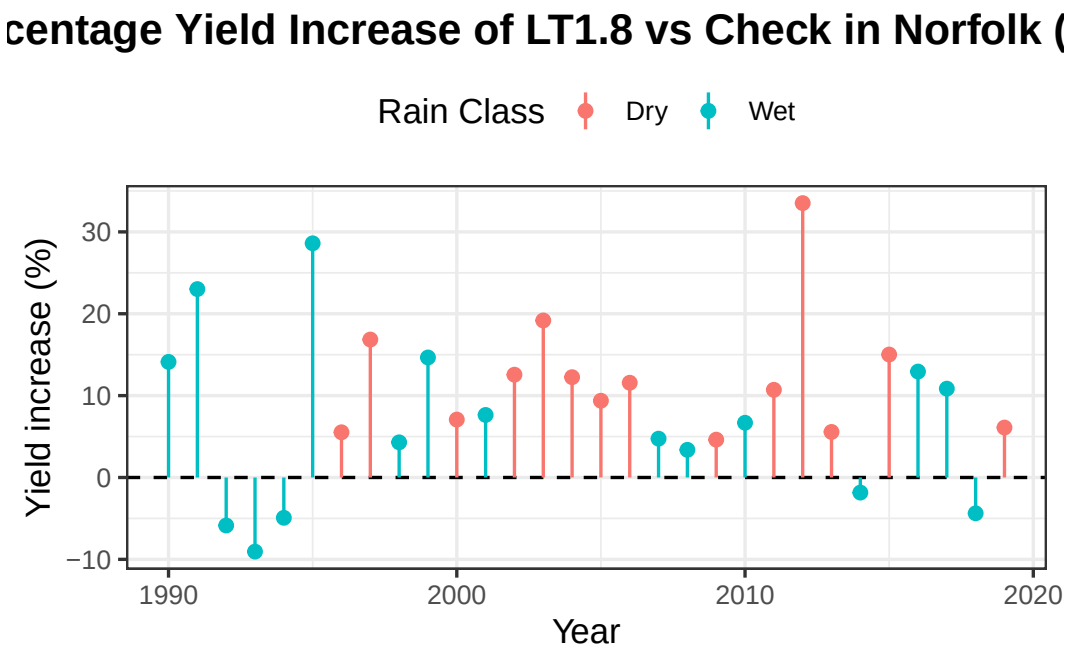
```r
analyze_site_lt18 <- function(site_name) {
  site_dat <- rainfed_data %>%
    filter(Location == site_name)

  by_year <- site_dat %>%
    mutate(geno = case_when(
      str_detect(Crop, regex("check", ignore_case = TRUE)) ~ "Check",
```

```
    str_detect(Crop, regex("LT[-_\\.]?1\\.8|\\b1\\.8\\b", ignore_case = TRUE)) ~ "LT1.8",
    TRUE ~ NA_character_
  )) %>%
  filter(!is.na(geno), !is.na(Yield_kgha)) %>%
  group_by(YEAR, geno) %>%
  summarise(yld = mean(Yield_kgha, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(
    names_from = geno,
    values_from = yld,
    values_fill = list(yld = NA_real_)
  ) %>%
  mutate(pct_increase = (`LT1.8` - Check) / Check * 100)

by_year_joined <- left_join(
  by_year,
  RAIN_ALL %>% dplyr::select(YEAR, Location, Rain.class) %>% filter(Location == site_name)
  by = "YEAR"
)

avg_check <- mean(by_year_joined$Check, na.rm = TRUE)
avg_lt18  <- mean(by_year_joined$`LT1.8`, na.rm = TRUE)
print(paste(site_name, "mean Check yield:", round(avg_check, 2)))
print(paste(site_name, "mean LT1.8 yield:", round(avg_lt18, 2)))

p <- ggplot(by_year_joined, aes(x = YEAR, y = pct_increase, colour = Rain.class)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_linerange(aes(ymin = 0, ymax = pct_increase)) +
  geom_point(size = 2) +
  labs(
    title = paste0("Percentage Yield Increase of LT1.8 vs Check in ", site_name, " (Rainfe
    x = "Year",
    y = "Yield increase (%)",
    colour = "Rain Class"
  ) +
  theme_bw(base_size = 13) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "top",
    strip.background = element_rect(fill = "grey90"),
    strip.text = element_text(face = "bold")
  )
```

```r
  output_dir <- "/Users/lucasromano/Desktop/Crop Modeling/Corn SSM/graphs, tables and diagram
  file_stub <- gsub(" ", "", site_name)

  emit_figure(
    plot_obj = p,
    caption = paste0(site_name, ": LT-1.8 vs Check (% yield increase by year, rain class)"),
    save_path = file.path(output_dir, paste0(file_stub, "_RFD_LT18_vs_Check.tiff")),
    width = 7,
    height = 4,
    dpi = 300,
    device = "tiff"
  )

  invisible(by_year_joined)
}

site_results <- list(
  Norfolk = analyze_site_lt18("Norfolk"),
  Jonesboro = analyze_site_lt18("Jonesboro"),
  Marianna = analyze_site_lt18("Marianna"),
  Stuttgart = analyze_site_lt18("Stuttgart"),
  `North Platte` = analyze_site_lt18("North Platte"),
  Lincoln = analyze_site_lt18("Lincoln")
)
```

```
[1] "Norfolk mean Check yield: 7252.53"
[1] "Norfolk mean LT1.8 yield: 7699.29"
```

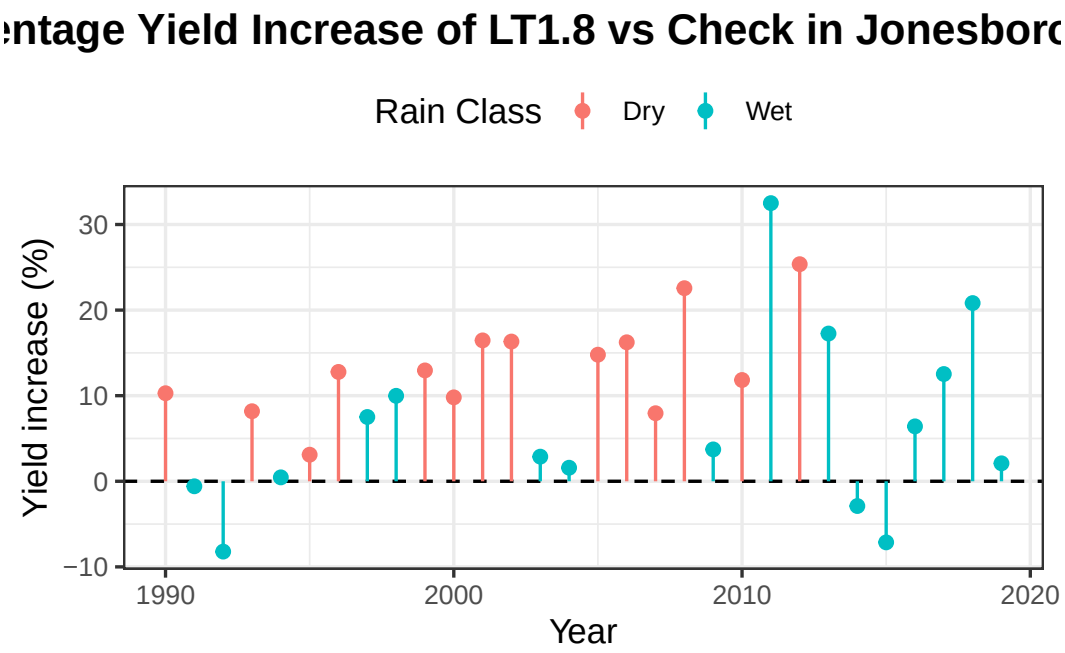
```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```




```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

```
[1] "Jonesboro mean Check yield: 7466.46"
[1] "Jonesboro mean LT1.8 yield: 7967.02"
```

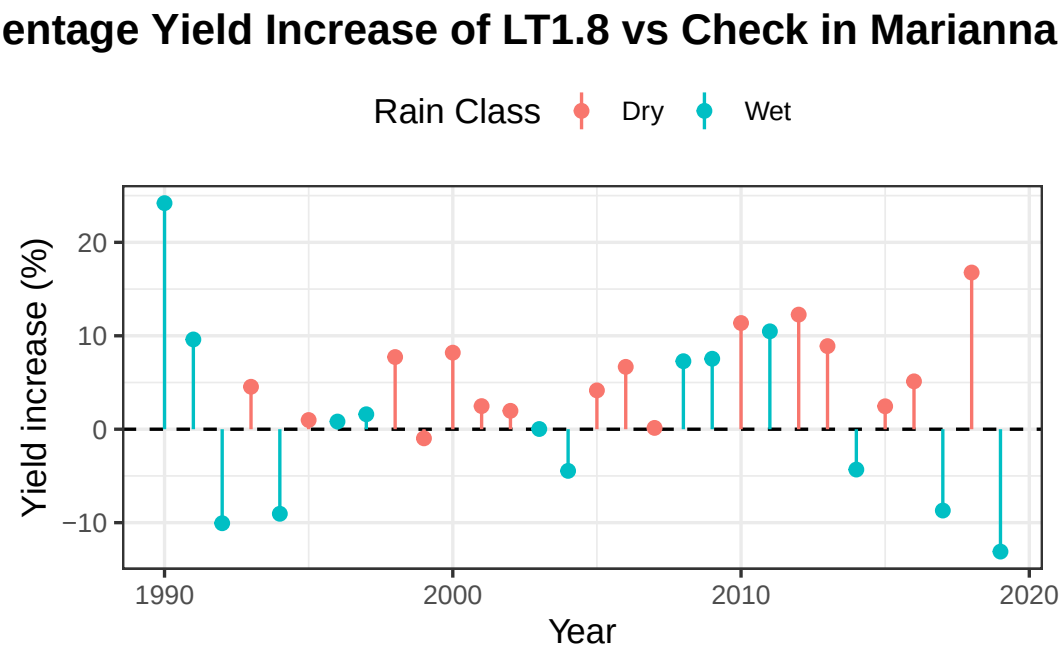
```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```




```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

```
[1] "Marianna mean Check yield: 8342.64"
[1] "Marianna mean LT1.8 yield: 8449.38"
```

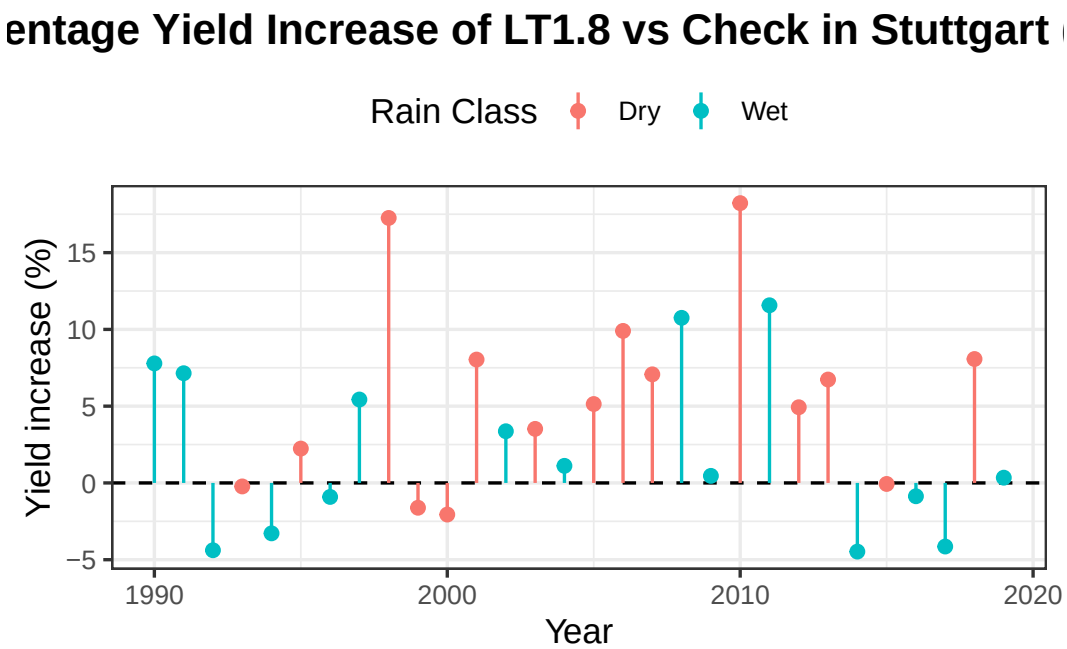
```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```




```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

```
[1] "Stuttgart mean Check yield: 6937.81"
[1] "Stuttgart mean LT1.8 yield: 7091.7"
```

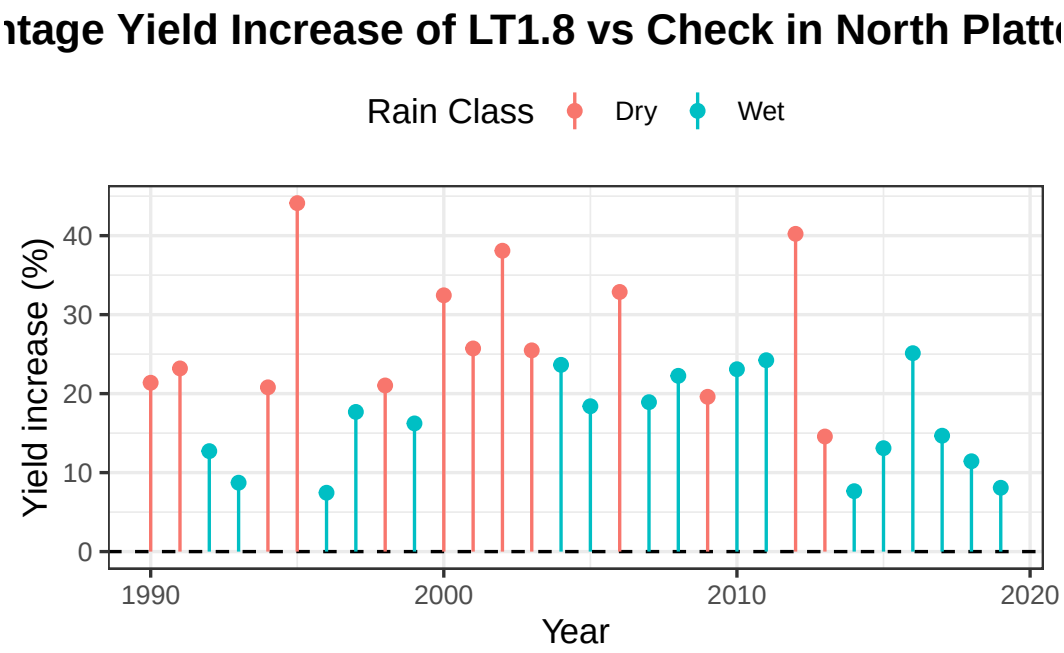
```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```




```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

```
[1] "North Platte mean Check yield: 6314.65"
[1] "North Platte mean LT1.8 yield: 7464.12"
```

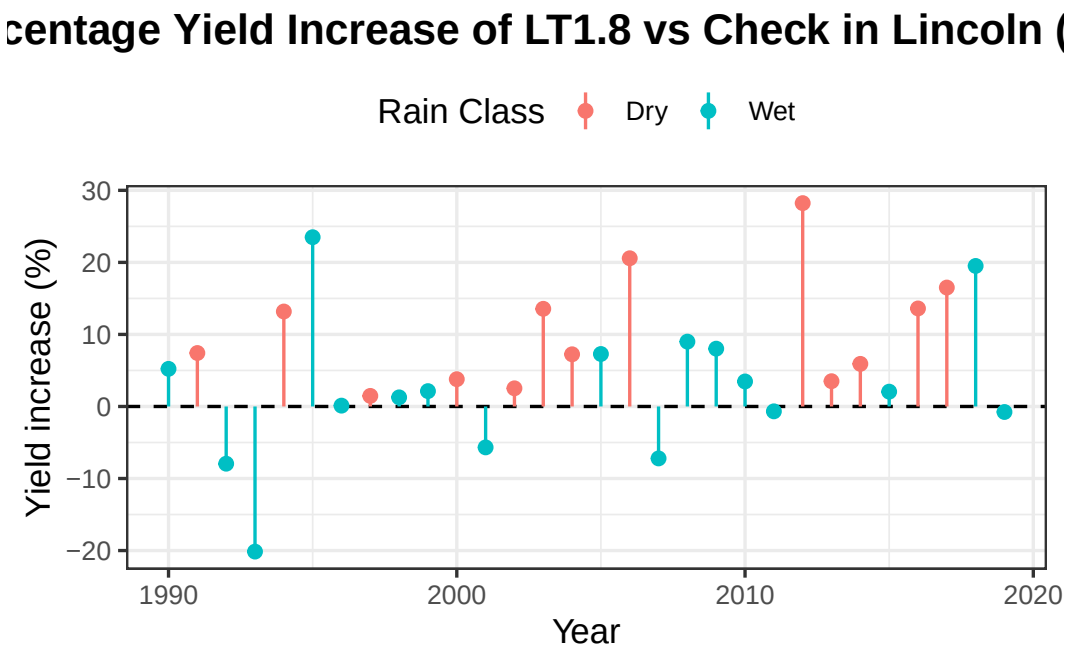
```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```




```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

```
[1] "Lincoln mean Check yield: 6186.22"
[1] "Lincoln mean LT1.8 yield: 6400.47"
```

```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```




```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

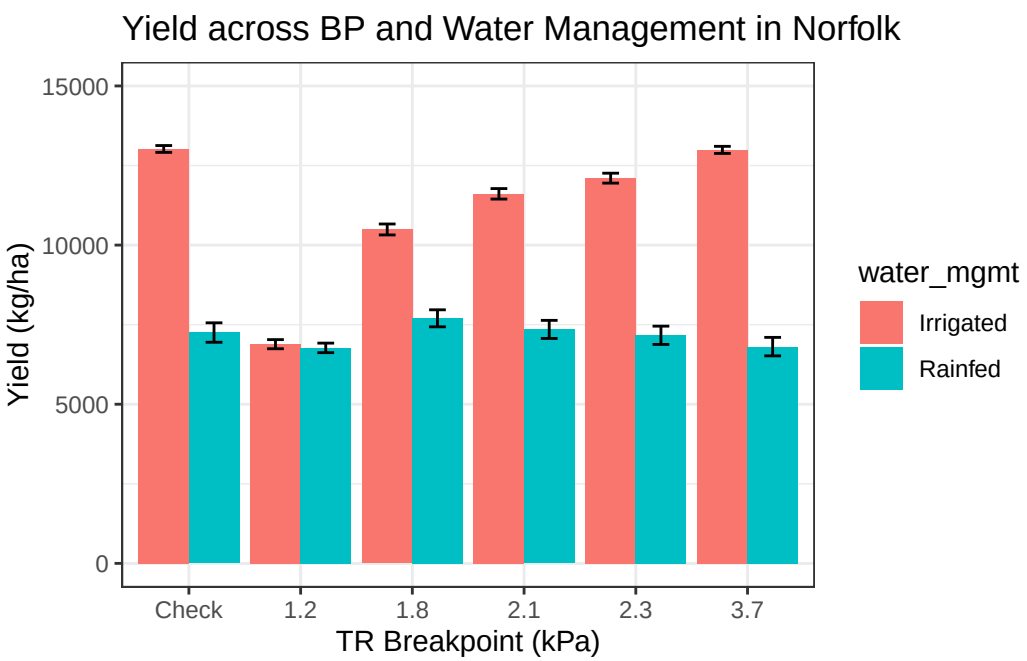
8) Plotting functions and utilities

```r
# ---- CROP LABEL MAPPING ----
# Simplify crop names in plot labels (e.g., "Maize-AR-check" → "Check")
crop_labels <- c(
  "Maize-AR-check" = "Check",
  "Maize-AR-LT-1.2" = "1.2",
  "Maize-AR-LT-1.8" = "1.8",
  "Maize-AR-LT-2.1" = "2.1",
  "Maize-AR-LT-2.3" = "2.3",
  "Maize-AR-LT-3.7" = "3.7",
  "Maize-NE-check" = "Check",
  "Maize-NE-LT-1.2" = "1.2",
  "Maize-NE-LT-1.8" = "1.8",
  "Maize-NE-LT-2.1" = "2.1",
  "Maize-NE-LT-2.3" = "2.3",
```

```r
  "Maize-NE-LT-3.7" = "3.7"
)
# ---- FUNCTION: Plot yield by crop × water management ----
# Generates bar chart with error bars (Standard Error) for a given location
output_dir <- "/Users/lucasromano/Desktop/Crop Modeling/Corn SSM/graphs, tables and diagrams"

plot_yield_summary <- function(location_data, location_name, filename) {
  # Calculate summary statistics (mean ± SE) by crop and water management
  summary_data <- location_data %>%
    group_by(Crop, water_mgmt) %>%
    summarise(
      mean_yield = mean(Yield_kgha, na.rm = TRUE),
      sd_yield = sd(Yield_kgha, na.rm = TRUE),
      n = n(),
      se_yield = sd_yield / sqrt(n),
      .groups = "drop"
    )

  # Create visualization with grouped bars and error bars
  p <- ggplot(summary_data, aes(x = Crop, y = mean_yield, fill = water_mgmt)) +
    geom_col(position = position_dodge(width = 0.9)) +             # Grouped columns
    geom_errorbar(                                                 # Standard error bars
      aes(ymin = mean_yield - se_yield, ymax = mean_yield + se_yield),
      position = position_dodge(width = 0.9),
      width = 0.3
    ) +
    scale_x_discrete(labels = crop_labels) +                      # Use short labels
    scale_y_continuous(limits = c(0, 15000)) +                 # Fixed y-axis range
    labs(
      title = paste("Yield across BP and Water Management in", location_name),
      x = "TR Breakpoint (kPa)",
      y = "Yield (kg/ha)"
    ) +
    theme_bw()

  # Save and display
  emit_figure(
    plot_obj = p,
    caption = paste("Yield across BP and Water Management in", location_name),
    save_path = file.path(output_dir, filename),
    width = 9,
    height = 6,
```

```
    dpi = 300,
    device = "tiff"
  )
}
# ---- EXAMPLE: Norfolk Yield Plot ----
plot_yield_summary(norfolk_data, "Norfolk", "YxBPxWM_Norfolk.tiff")
```

```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

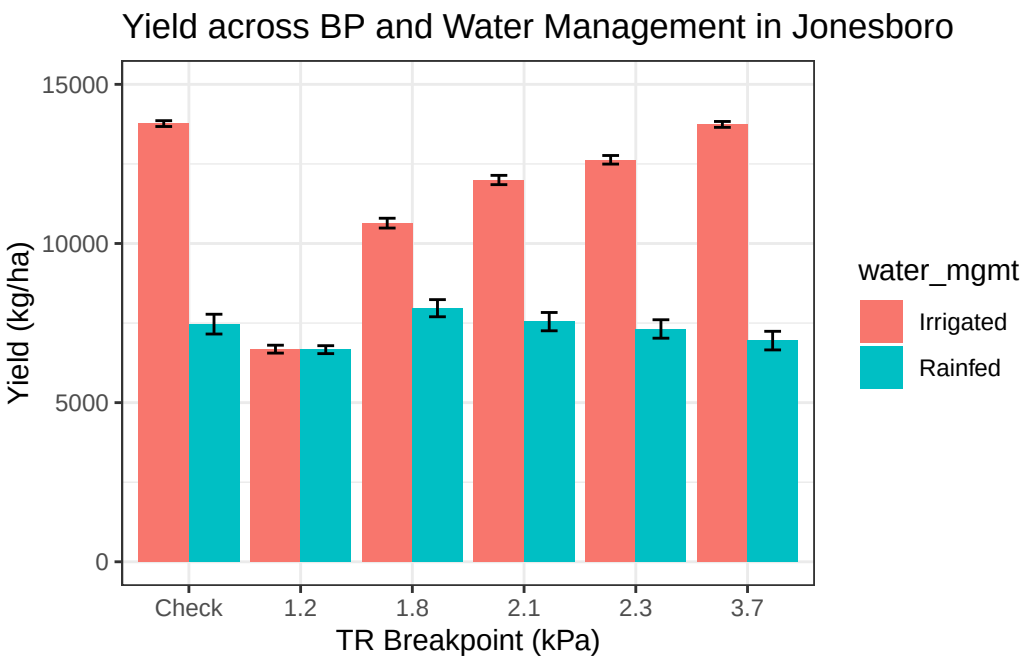



```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

```
plot_yield_summary(jonesboro_data, "Jonesboro", "YxBPxWM_Jonesboro.tiff")
```

```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
```

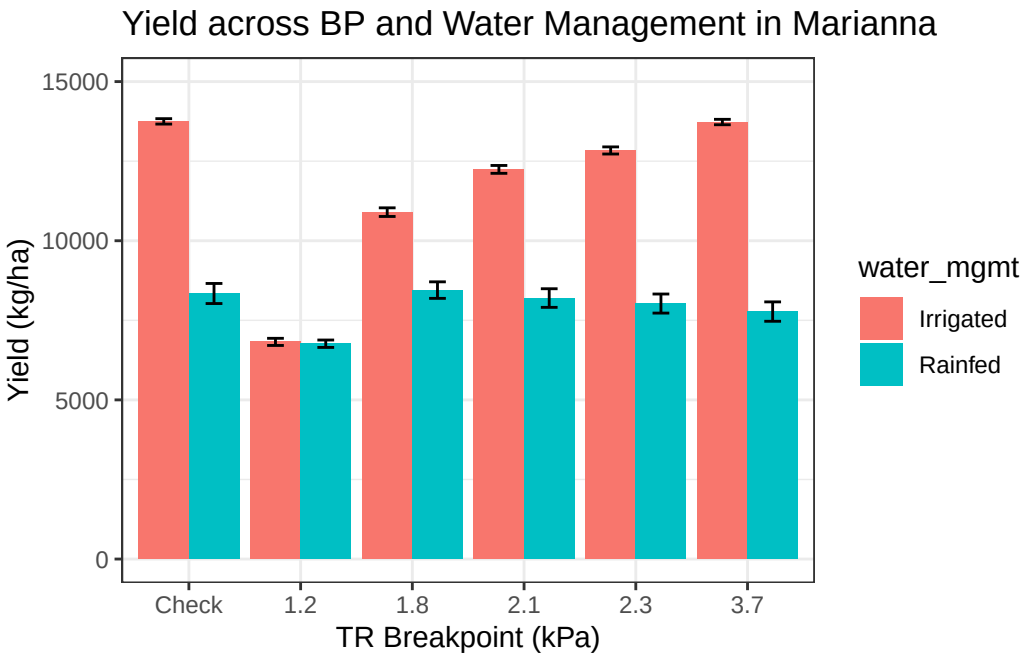
```
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```




```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

```
plot_yield_summary(marianna_data, "Marianna", "YxBPxWM_Marianna.tiff")
```

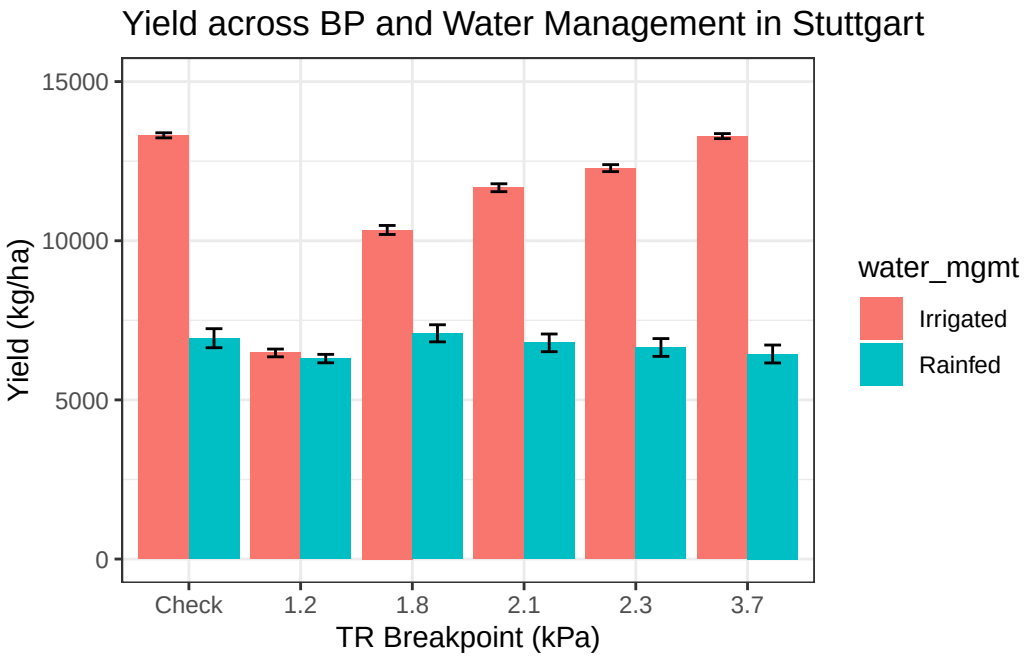
```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```




```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

```
plot_yield_summary(stuttgart_data, "Stuttgart", "YxBPxWM_Stuttgart.tiff")
```

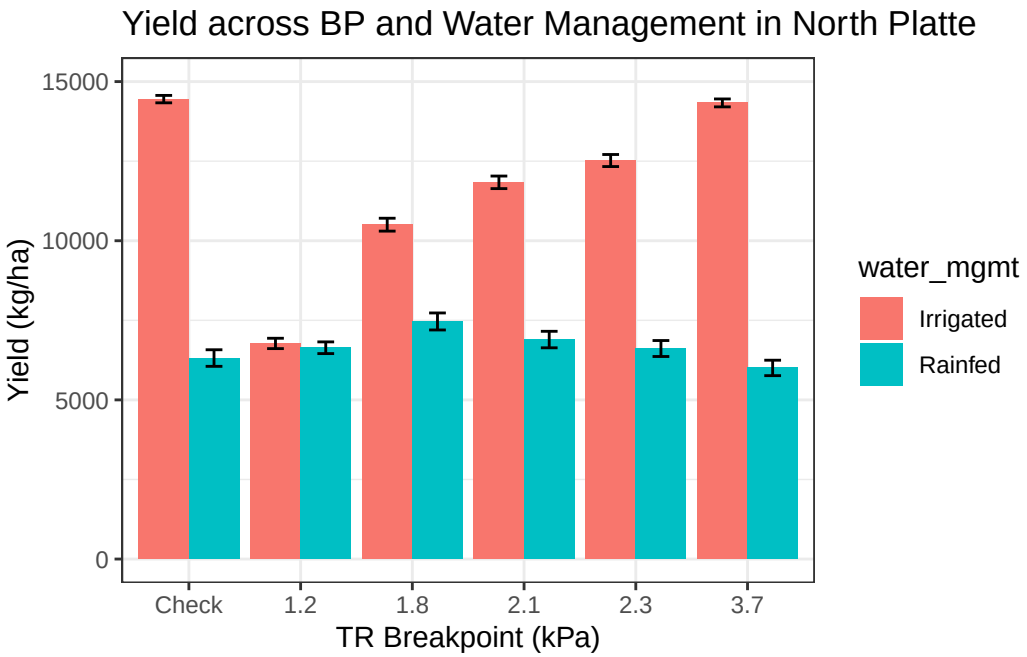
```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```




```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

```
plot_yield_summary(north_platte_data, "North Platte", "YxBPxWM_NorthPlatte.tiff")
```

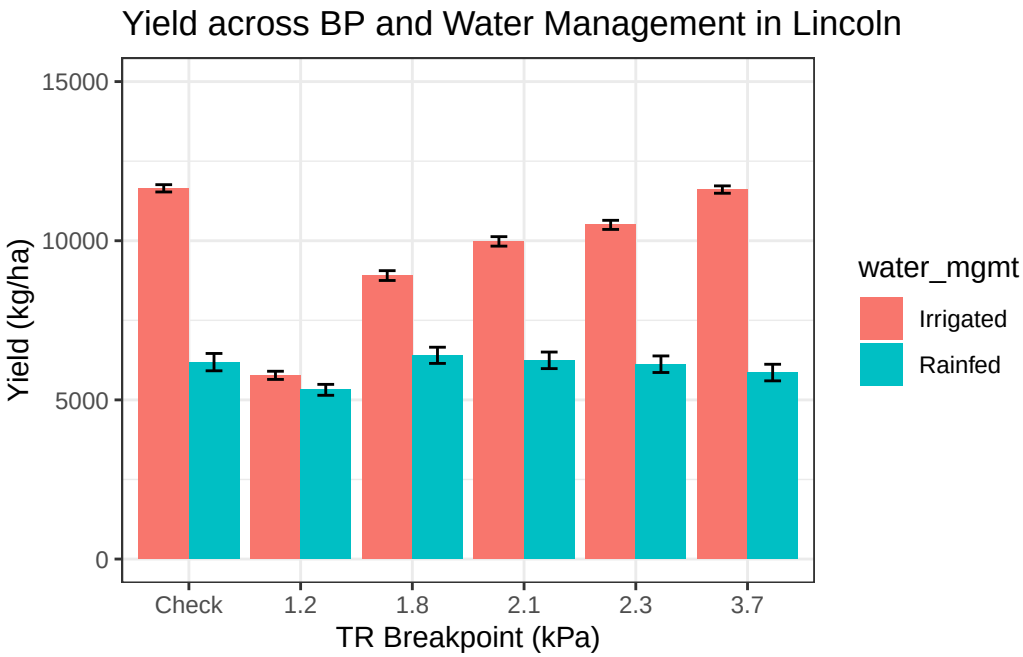
```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```




```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

```
plot_yield_summary(lincoln_data, "Lincoln", "YxBPxWM_Lincoln.tiff")
```

```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```




```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

9) Water use efficiency (WUE) analysis

Analyze yield response to cumulative transpiration (CTR) for different regions.

—- WUE: shared utilities —-

```r
library(ggplot2)
library(ggpmisc)
```

```
Loading required package: ggpp

Registered S3 methods overwritten by 'ggpp':
  method                  from
  heightDetails.titleGrob ggplot2
  widthDetails.titleGrob  ggplot2
```

```
Attaching package: 'ggpp'

The following object is masked from 'package:ggplot2':

    annotate
```

```r
build_wue_plot <- function(data, water_mode, crop_ids, plot_title, color_var = NULL) {
  plot_data <- data %>%
    filter(
      water_mgmt == water_mode,
      Crop %in% crop_ids
    )

  if (is.null(color_var)) {
    p <- ggplot(plot_data, aes(x = CTR, y = Yield_kgha))
  } else {
    p <- ggplot(plot_data, aes(x = CTR, y = Yield_kgha, color = .data[[color_var]]))
  }

  p +
    geom_point(alpha = 0.7, size = 2) +
    geom_smooth(method = "lm", se = FALSE, linetype = "solid", linewidth = 1.1) +
    stat_poly_eq(
      aes(label = paste(..eq.label.., ..rr.label.., sep = "~~~")),
      formula = y ~ x,
      parse = TRUE,
      label.x.npc = "left",
      label.y.npc = 0.95,
      size = 3
    ) +
    facet_wrap(~Crop, scales = "fixed") +
    labs(
      title = plot_title,
      x = "Cumulative Transpiration (mm)",
      y = "Grain Yield (kg ha-1)"
    ) +
    theme_bw(base_size = 13) +
    theme(
      plot.title = element_text(hjust = 0.5, face = "bold"),
      legend.position = "top",
      strip.background = element_rect(fill = "grey90"),
```

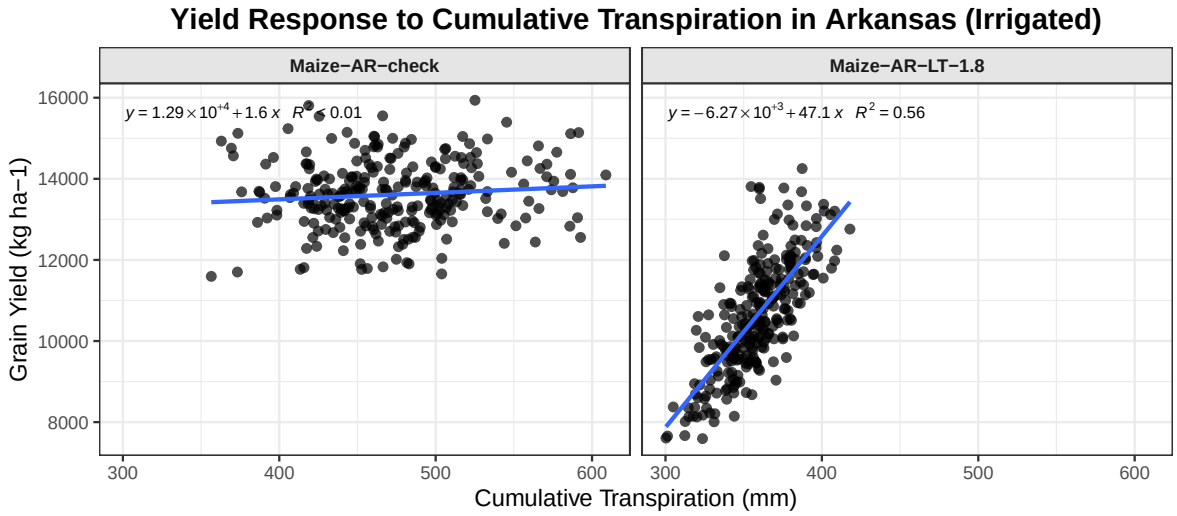
```
      strip.text = element_text(face = "bold")
    )
}

save_wue_plot <- function(plot_obj, filename) {
  ggsave(
    filename = filename,
    plot = plot_obj,
    width = 9,
    height = 4,
    dpi = 300,
    device = "tiff"
  )
}
```

—- WUE: all scenarios using one loop —-

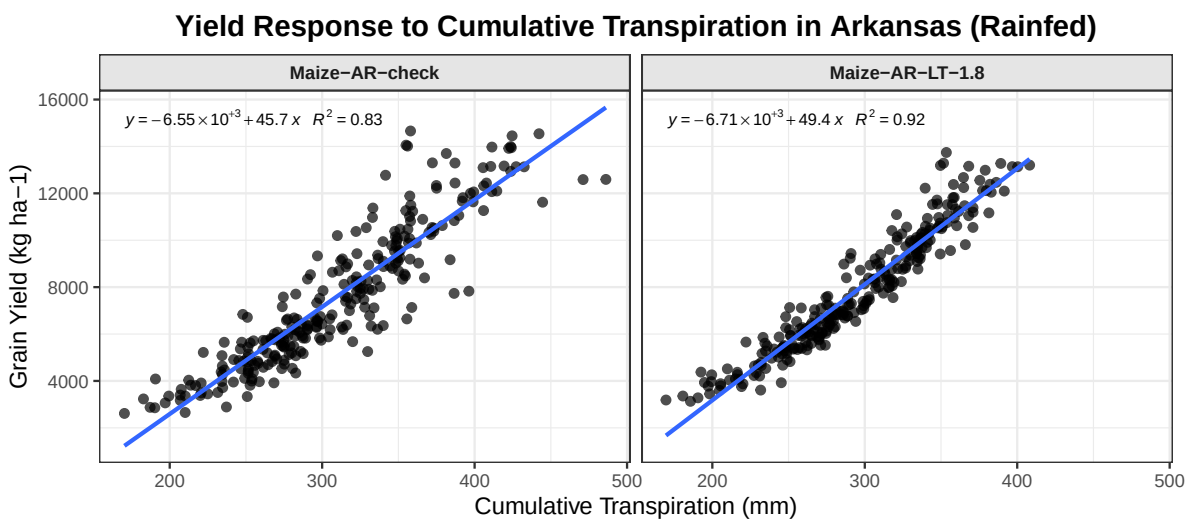
```
wue_scenarios <- list(
  list(
    obj_name = "WUE_AR_Irrigated",
    data_obj = corn_sim_clean,
    water_mode = "Irrigated",
    crop_ids = c("Maize-AR-check", "Maize-AR-LT-1.8"),
    title = "Yield Response to Cumulative Transpiration in Arkansas (Irrigated)",
    color_var = NULL,
    file_name = "WUE_AR_Irrigated.tiff"
  ),
  list(
    obj_name = "WUE_AR_Rainfed",
    data_obj = corn_sim_clean,
    water_mode = "Rainfed",
    crop_ids = c("Maize-AR-check", "Maize-AR-LT-1.8"),
    title = "Yield Response to Cumulative Transpiration in Arkansas (Rainfed)",
    color_var = NULL,
    file_name = "WUE_AR_Rainfed.tiff"
  ),
  list(
    obj_name = "WUE_NE_Rainfed",
    data_obj = corn_sim_clean,
```

```r
    water_mode = "Rainfed",
    crop_ids = c("Maize-NE-check", "Maize-NE-LT-1.8"),
    title = "Yield Response to Cumulative Transpiration in Nebraska (Rainfed)",
    color_var = NULL,
    file_name = "WUE_NE_Rainfed.tiff"
  ),
  list(
    obj_name = "WUE_NE_Irrigated",
    data_obj = corn_sim_clean,
    water_mode = "Irrigated",
    crop_ids = c("Maize-NE-check", "Maize-NE-LT-1.8"),
    title = "Yield Response to Cumulative Transpiration in Nebraska (Irrigated)",
    color_var = NULL,
    file_name = "WUE_NE_Irrigated.tiff"
  )
)

output_dir <- "/Users/lucasromano/Desktop/Crop Modeling/Corn SSM/graphs, tables and diagrams"

for (scenario in wue_scenarios) {
  wue_plot <- build_wue_plot(
    data = scenario$data_obj,
    water_mode = scenario$water_mode,
    crop_ids = scenario$crop_ids,
    plot_title = scenario$title,
    color_var = scenario$color_var
  )
  assign(scenario$obj_name, wue_plot)
  save_wue_plot(wue_plot, file.path(output_dir, scenario$file_name))
  print(wue_plot)
}
```

```
Warning in stat_poly_eq(aes(label = paste(..eq.label.., ..rr.label.., sep =
"~~~")), : Ignoring unknown parameters: `label.x.npc` and `label.y.npc`

Warning: The dot-dot notation (`..eq.label..`) was deprecated in ggplot2 3.4.0.
i Please use `after_stat(eq.label)` instead.

`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```

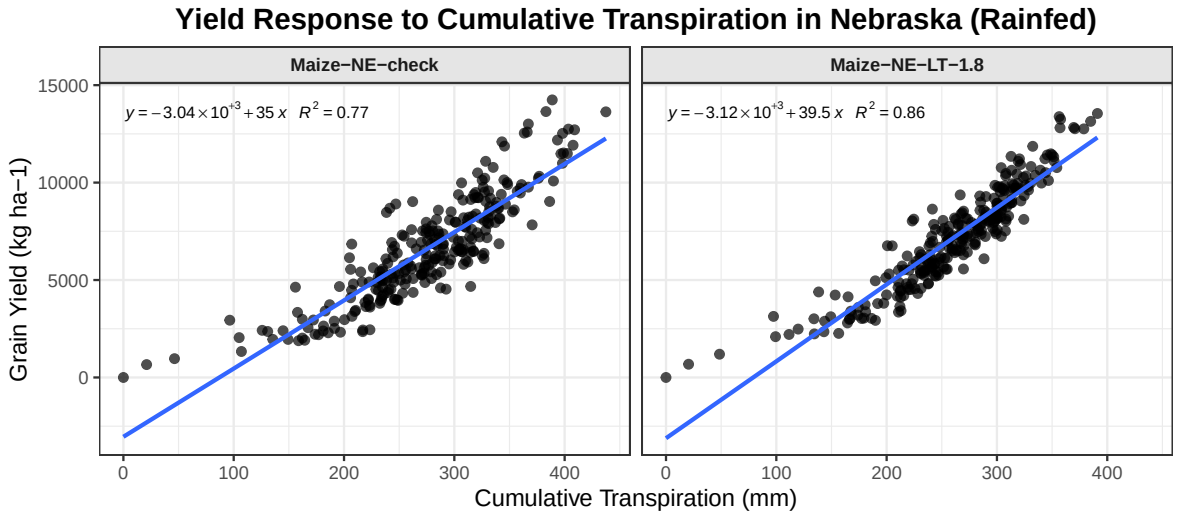



```
Warning in stat_poly_eq(aes(label = paste(..eq.label.., ..rr.label.., sep =
"~~~")), : Ignoring unknown parameters: `label.x.npc` and `label.y.npc`

`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```



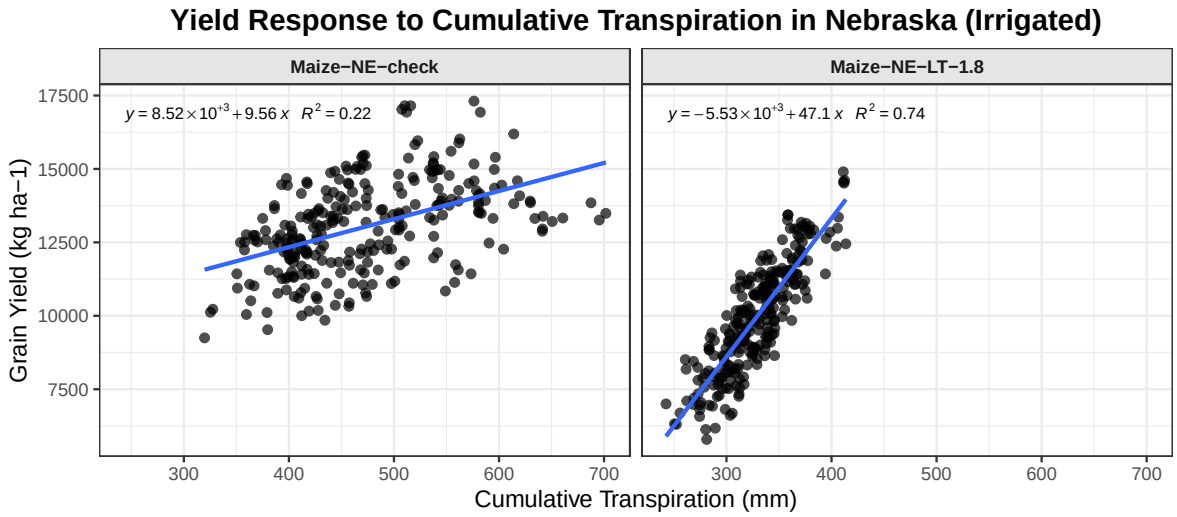

```
Warning in stat_poly_eq(aes(label = paste(..eq.label.., ..rr.label.., sep =
"~~~")), : Ignoring unknown parameters: `label.x.npc` and `label.y.npc`
```

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```




```
Warning in stat_poly_eq(aes(label = paste(..eq.label.., ..rr.label.., sep =
"~~~")), : Ignoring unknown parameters: `label.x.npc` and `label.y.npc`
```

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```



```r
# Reset caption option after loop
knitr::opts_current$set(fig.cap = NULL)
```

```
Warning in set2(resolve(...)): The object is read-only and cannot be modified.
If you have to modify it for a legitimate reason, call the method $lock(FALSE)
on the object before $set(). Using $lock(FALSE) to modify the object will be
enforced in future versions of knitr and this warning will become an error.
```

10) Yield stability analysis

Compare year-to-year variability (CV) in yields for Check vs LT genotypes.

```r
library(dplyr)
library(ggplot2)

build_stability_data <- function(data, water_mode, g1, g2, label_map = NULL) {
  df_year <- data %>%
    filter(water_mgmt == water_mode, Crop %in% c(g1, g2)) %>%
    group_by(Crop, YEAR) %>%
    summarise(y = mean(Yield_kgha, na.rm = TRUE), .groups = "drop")

  if (is.null(label_map)) {
    df_year <- df_year %>% mutate(Crop_pretty = Crop)
  } else {
    df_year <- df_year %>% mutate(Crop_pretty = as.character(label_map[Crop]))
  }

  stab <- df_year %>%
    group_by(Crop, Crop_pretty) %>%
    summarise(
      mean_y = mean(y),
      sd_y = sd(y),
      cv = 100 * sd_y / mean_y,
      .groups = "drop"
    )

  subtxt <- stab %>%
    mutate(lbl = sprintf("%s: mean = %.0f kg/ha, CV = %.1f%%", Crop_pretty, mean_y, cv)) %>%
    pull(lbl) %>%
    paste(collapse = "   |   ")
```

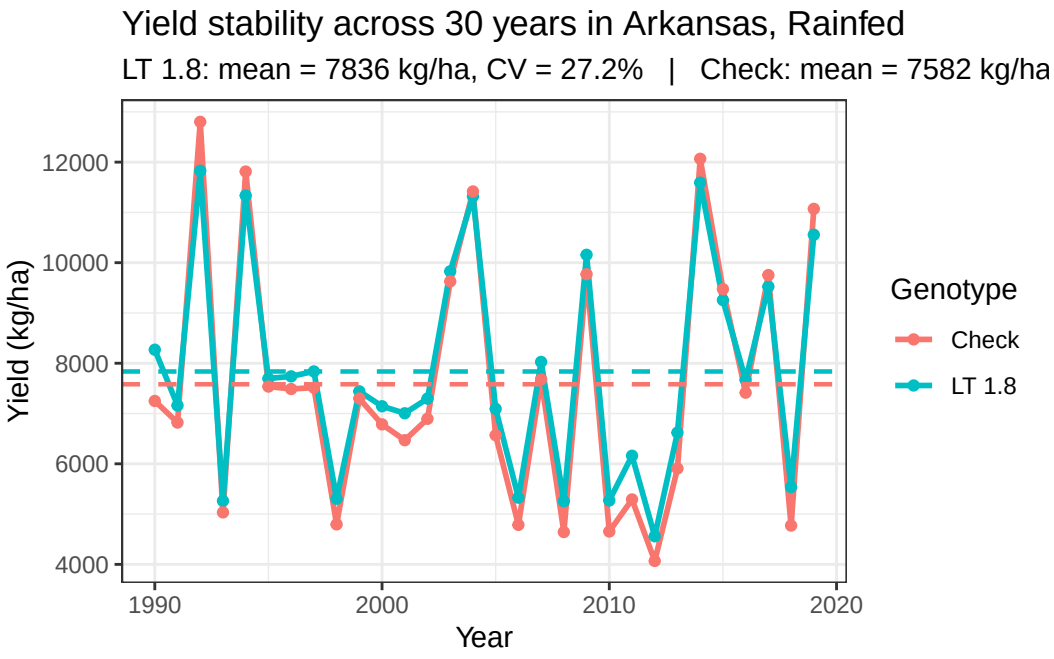
```r
  list(df_year = df_year, stab = stab, subtxt = subtxt)
}

plot_stability_basic <- function(stab_obj, title_text) {
  ggplot(stab_obj$df_year, aes(x = YEAR, y = y, color = Crop_pretty, group = Crop_pretty)) +
    geom_line(linewidth = 1) +
    geom_point() +
    geom_hline(
      data = stab_obj$stab,
      aes(yintercept = mean_y, color = Crop_pretty),
      linetype = "dashed",
      linewidth = 0.8,
      show.legend = FALSE
    ) +
    labs(
      title = title_text,
      subtitle = stab_obj$subtxt,
      x = "Year",
      y = "Yield (kg/ha)",
      color = "Genotype"
    ) +
    theme_bw()
}

plot_stability_poster <- function(stab_obj, title_text, palette) {
  ggplot(
    stab_obj$df_year,
    aes(x = YEAR, y = y, color = Crop_pretty, group = Crop_pretty)
  ) +
    geom_line(linewidth = 1.4) +
    geom_point(size = 3, stroke = 1, fill = "white", shape = 21, show.legend = FALSE) +
    geom_hline(
      data = stab_obj$stab,
      aes(yintercept = mean_y, color = Crop_pretty),
      linetype = "dashed",
      linewidth = 1,
      show.legend = FALSE
    ) +
    scale_color_manual(values = palette, name = "Genotype") +
    scale_y_continuous(labels = function(x) format(x, big.mark = ",", scientific = FALSE)) +
    labs(
      title = title_text,
```

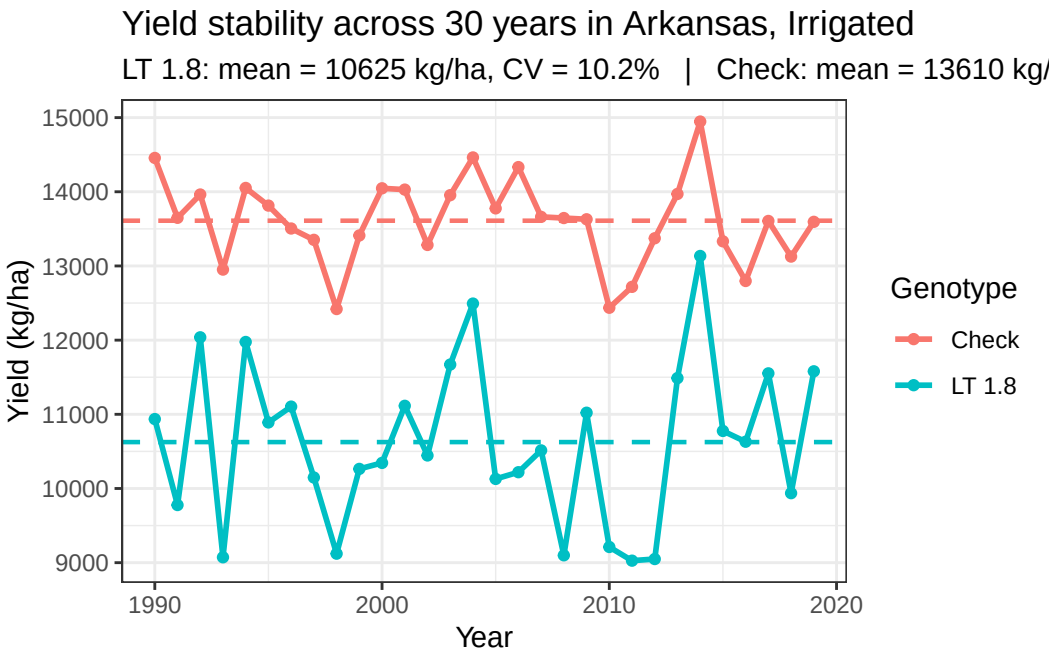
```
      subtitle = stab_obj$subtxt,
      x = "Year",
      y = "Yield (kg ha⁻¹)",
      color = "Genotype"
    ) +
    theme_minimal(base_size = 15, base_family = "Helvetica") +
    theme(
      plot.title = element_text(size = 20, face = "bold", hjust = 0.5, lineheight = 1.1,
      margin = margin(b = 6)),
      plot.subtitle = element_text(size = 14, hjust = 0.5, margin = margin(b = 12)),
      axis.title = element_text(size = 16, face = "bold"),
      axis.text = element_text(size = 14),
      axis.line.x = element_line(linewidth = 0.8, color = "black"),
      axis.line.y = element_line(linewidth = 0.8, color = "black"),
      panel.grid.major.y = element_line(color = "gray85", linewidth = 0.6),
      panel.grid.minor.y = element_blank(),
      panel.grid.major.x = element_blank(),
      panel.grid.minor.x = element_blank(),
      legend.position = "top",
      legend.title = element_text(size = 14, face = "bold"),
      legend.text = element_text(size = 13),
      plot.margin = margin(15, 20, 15, 20)
    )
}

save_stability_plot <- function(plot_obj, file_name, width, height, dpi) {
  ggsave(
    filename = file.path(output_dir, file_name),
    plot = plot_obj,
    width = width,
    height = height,
    dpi = dpi,
    compression = "lzw",
    device = "tiff"
  )
}
```

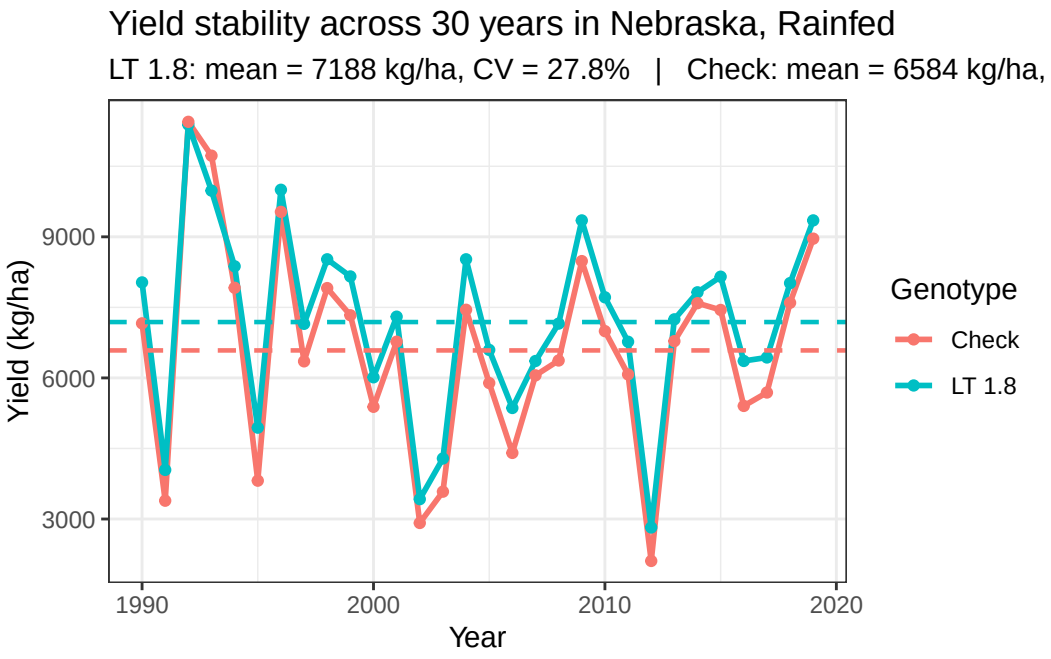
```
stability_scenarios <- list(
  list(
    obj_name = "Ystab_AR_rfd",
    water_mode = "Rainfed",
    g1 = "Maize-AR-check",
```

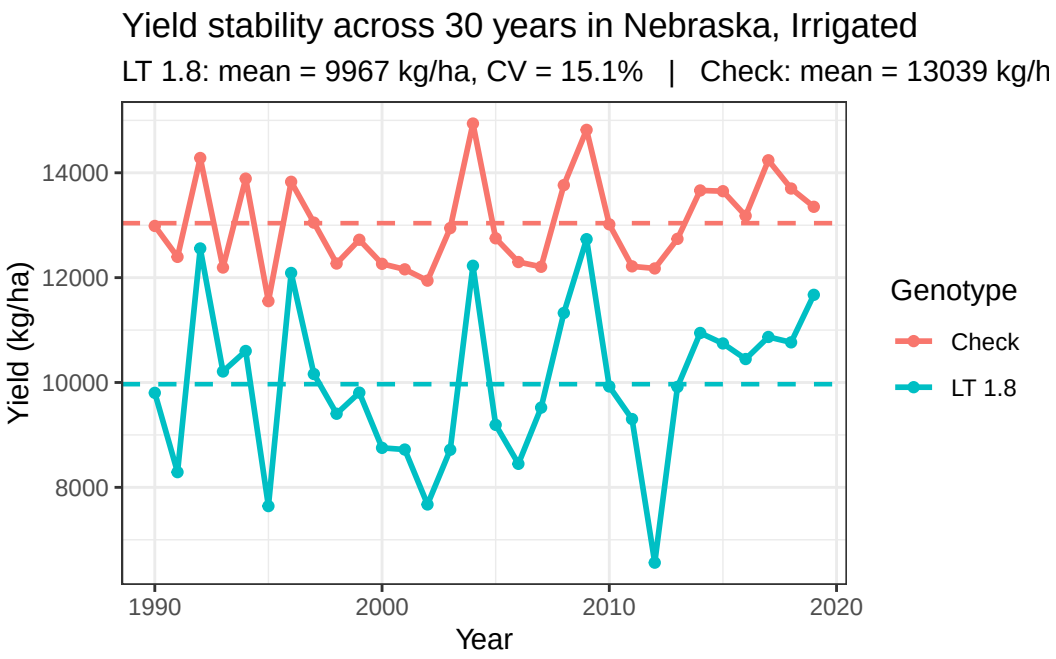
```
    g2 = "Maize-AR-LT-1.8",
    label_map = c("Maize-AR-check" = "Check", "Maize-AR-LT-1.8" = "LT 1.8"),
    style = "basic",
    palette = NULL,
    title = "Yield stability across 30 years in Arkansas, Rainfed",
    file_name = "Ystab_AR_rfd_poster.tiff",
    width = 10,
    height = 6,
    dpi = 600
  ),
  list(
    obj_name = "Ystab_AR_irri",
    water_mode = "Irrigated",
    g1 = "Maize-AR-check",
    g2 = "Maize-AR-LT-1.8",
    label_map = c("Maize-AR-check" = "Check", "Maize-AR-LT-1.8" = "LT 1.8"),
    style = "basic",
    palette = NULL,
    title = "Yield stability across 30 years in Arkansas, Irrigated",
    file_name = "Ystab_AR_irri.tiff",
    width = 9,
    height = 4,
    dpi = 300
  ),
  list(
    obj_name = "Ystab_NE_rfd",
    water_mode = "Rainfed",
    g1 = "Maize-NE-check",
    g2 = "Maize-NE-LT-1.8",
    label_map = c("Maize-NE-check" = "Check", "Maize-NE-LT-1.8" = "LT 1.8"),
    style = "basic",
    palette = NULL,
    title = "Yield stability across 30 years in Nebraska, Rainfed",
    file_name = "Ystab_NE_rfd.tiff",
    width = 9,
    height = 4,
    dpi = 300
  ),
  list(
    obj_name = "Ystab_NE_irri",
    water_mode = "Irrigated",
    g1 = "Maize-NE-check",
```

```r
    g2 = "Maize-NE-LT-1.8",
    label_map = c("Maize-NE-check" = "Check", "Maize-NE-LT-1.8" = "LT 1.8"),
    style = "basic",
    palette = NULL,
    title = "Yield stability across 30 years in Nebraska, Irrigated",
    file_name = "Ystab_NE_irri.tiff",
    width = 9,
    height = 4,
    dpi = 300
  )
)

for (scenario in stability_scenarios) {
  stab_obj <- build_stability_data(
    data = corn_sim_clean,
    water_mode = scenario$water_mode,
    g1 = scenario$g1,
    g2 = scenario$g2,
    label_map = scenario$label_map
  )

  stab_plot <- if (scenario$style == "poster") {
    plot_stability_poster(stab_obj, scenario$title, scenario$palette)
  } else {
    plot_stability_basic(stab_obj, scenario$title)
  }

  assign(scenario$obj_name, stab_plot)
  save_stability_plot(
    plot_obj = stab_plot,
    file_name = scenario$file_name,
    width = scenario$width,
    height = scenario$height,
    dpi = scenario$dpi
  )

  print(stab_plot)
}
```


Yield stability across 30 years in Arkansas, Rainfed
LT 1.8: mean = 7836 kg/ha, CV = 27.2% | Check: mean = 7582 kg/ha
Yield (kg/ha)
12000
10000
8000
6000
4000
1990
2000
2010
2020
Year
Genotype
Check
LT 1.8


Yield stability across 30 years in Arkansas, Irrigated
LT 1.8: mean = 10625 kg/ha, CV = 10.2% | Check: mean = 13610 kg/
Yield (kg/ha)
15000
14000
13000
12000
11000
10000
9000
1990
2000
2010
2020
Year
Genotype
Check
LT 1.8


Yield stability across 30 years in Nebraska, Rainfed
LT 1.8: mean = 7188 kg/ha, CV = 27.8% | Check: mean = 6584 kg/ha,
Yield (kg/ha)
9000
6000
3000
1990
2000
2010
2020
Year
Genotype
Check
LT 1.8


Yield stability across 30 years in Nebraska, Irrigated
LT 1.8: mean = 9967 kg/ha, CV = 15.1% | Check: mean = 13039 kg/h
Yield (kg/ha)
14000
12000
10000
8000
1990
2000
2010
2020
Year
Genotype
Check
LT 1.8